

Smart Waste Collection and Monitoring System

Introduction

The Smart Waste Collection and Monitoring System's software design, system architecture, database structure, process models, feasibility study, and validation techniques are presented in this document. The SDAD describes how the system will work and how the suggested solution tackles local communities' waste management issues.

Process Architecture

The Administrator, Waste Collection Staff, and Residents are the three main external parties that the system communicates with. User management, schedule management, complaint management, reporting, and the notification system are among the system's primary functions. User data, collection schedules, complaint reports, notifications, and performance reports are all part of the system's data flows.

User Account Management, Waste Collection Scheduling, Complaint Handling, Monitoring and Reporting, and Notification Processing are the five main processes shown in the Level-0 Data Flow Diagram. These procedures show how data flows between users and the database and serve as a representation of the system's essential features.

User registration, login verification, and profile updating are all part of the User Management procedure. Staff assignment, schedule generation, and schedule modifications are all handled by the Schedule Management process. Complaint management enables personnel to address problems, administrators to analyze complaints, and residents to file complaints. Tracking the status of collections, creating reports, and presenting analytics for administrative needs are all part of monitoring and reporting.

Data Architecture

Users, Residents, Staff, Schedules, Complaints, Notifications, and Reports are some of the main entities that make up the system database. To support the system's functions and operations, these entities are connected.

User_id, full_name, email, password, and role are among the fields included in the Users table. Schedule_id, collection_date, area, and assigned_staff are all included in the Schedules table. Complaint_id, resident_id, complaint_text, status, and date_submitted are all stored in the Complaints table. These characteristics guarantee that system data is properly organized and retrieved.

The structure of the database relationships is one-to-many. Administrators can oversee numerous users within the system, staff members can manage various schedules, and residents can file multiple complaints.

In order to avoid redundancy, maintain appropriate dependency structures, and remove duplicate data, the database design adheres to the Third Normal Form (3NF). The efficiency and consistency of databases are enhanced by this normalization procedure.

TELOS Feasibility Study

The project's necessary technologies are available and appropriate for implementation at the BSIT level. For development and version control, the system makes use of HTML, CSS, Bootstrap, JavaScript, PHP, MySQL, and GitHub.

By using digital scheduling, automating reporting procedures, and cutting down on paperwork, the initiative lowers operating expenses. Utilizing open-source technologies also lowers development costs.

To guarantee lawful and responsible system operation, the system conforms with data privacy principles, ethical data handling methods, and approved user access policies.

Residents, waste collection workers, and barangay administration will find the system easy to use. To use the platform efficiently, very little technical knowledge is needed.

By applying Software Development Life Cycle (SDLC) approaches and allocating tasks appropriately, the project can be finished within the academic semester.

UI/UX Blueprint

To create a clear and effective user experience, the UI design adheres to the concepts of simplicity, accessibility, mobile responsiveness, and a theme inspired by sustainability.

The Landing Page offers login or registration options, an explanation of the system, and SDG branding. Users can view schedules, file complaints, and get alerts through the Resident Dashboard. The Staff Dashboard provides allocated routes, collection updates, and complaint answers. Analytics charts, user management tools, and monitoring reports are all included in the Administrator Dashboard.

Validation Plan

The validation and testing strategy guarantees that the Smart Waste Collection and Monitoring System meets all standards and operates as intended. Every system need, such as user login validation, schedule creation testing, complaint submission testing, and report generation testing, is matched with the proper test cases using a traceability matrix. Various testing techniques are used to assess the system's dependability and quality. Functional testing confirms that every feature and module performs as intended. Usability testing guarantees that administrators, employees, and residents can easily traverse the interface. Performance testing assesses the system's overall effectiveness, stability, and response time under typical usage circumstances.

Conclusion

A sustainable and effective digital solution for enhancing community trash management operations is offered by the Smart trash Collection and Monitoring System. Using the fundamental ideas of System Analysis and Design, the suggested architecture, process design, database structure, and validation techniques guarantee that the system is in line with SDG 11 objectives. The SDAD serves as the basis for upcoming implementation and prototype development and shows the project's operational and technical viability.